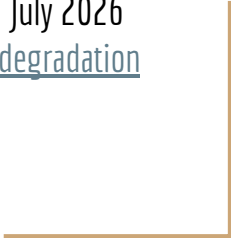




Early Prediction of Lithium-Ion Battery End-of-Life

GNEM Research Fellow · Levon Lau · July 2026
github.com/levonl22/gnem-battery-degradation



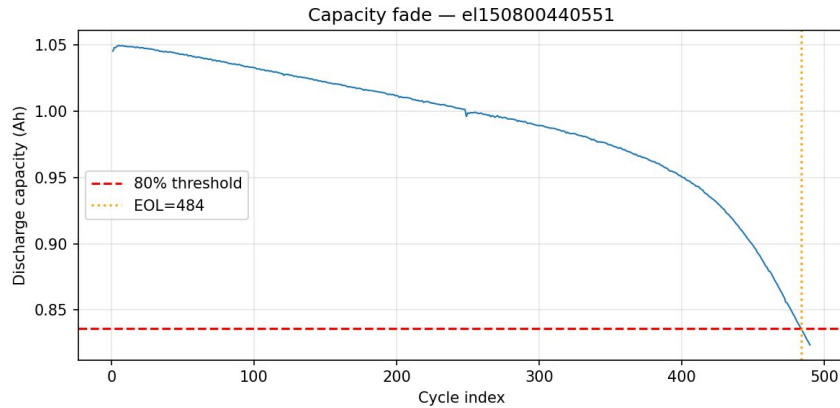
Problem & motivation

- Lithium-ion cells lose capacity over cycling; full life tests can run **hundreds to thousands of cycles**
- **Goal:** predict **end-of-life (EOL)** from **early-cycle** measurements before obvious fade
- **EOL definition:** first cycle where discharge capacity < **80%** of initial capacity (`cycle_index ≥ 1`)
- **Dataset:** MIT-Stanford-Toyota fast-charging cohort (Severson et al., 2019)
- **Approach:** reproducible Python pipeline → hand-crafted features + tree models vs GRU sequence model → monotonic SOH check → early-cycle window ablation

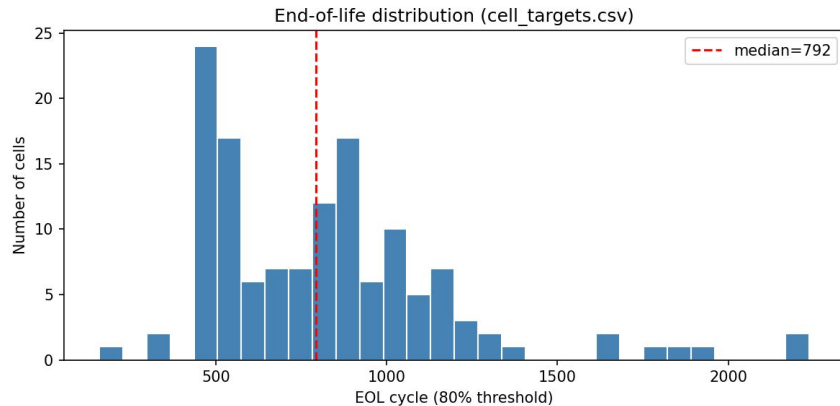
Dataset & pipeline

- **Raw:** 140 JSON files (`FastCharge*_structure.json`) from MATR.io
- **Cleaned:** **134 unique cells** (duplicate barcodes → longest run kept; one partial cell dropped)
- **Grain:** about **111k** per-cycle rows in `cycle_summary.csv`; **44** cell-level features in `cell_features.csv`
- **Split:** cell-level **94 / 20 / 20** train · val · test (`random_state = 42`) — never random row splits
- **Pipeline:** notebooks `01–10` · processed CSVs · metrics JSON · figures in `results/`

Labels — capacity fade & cycle-life spread

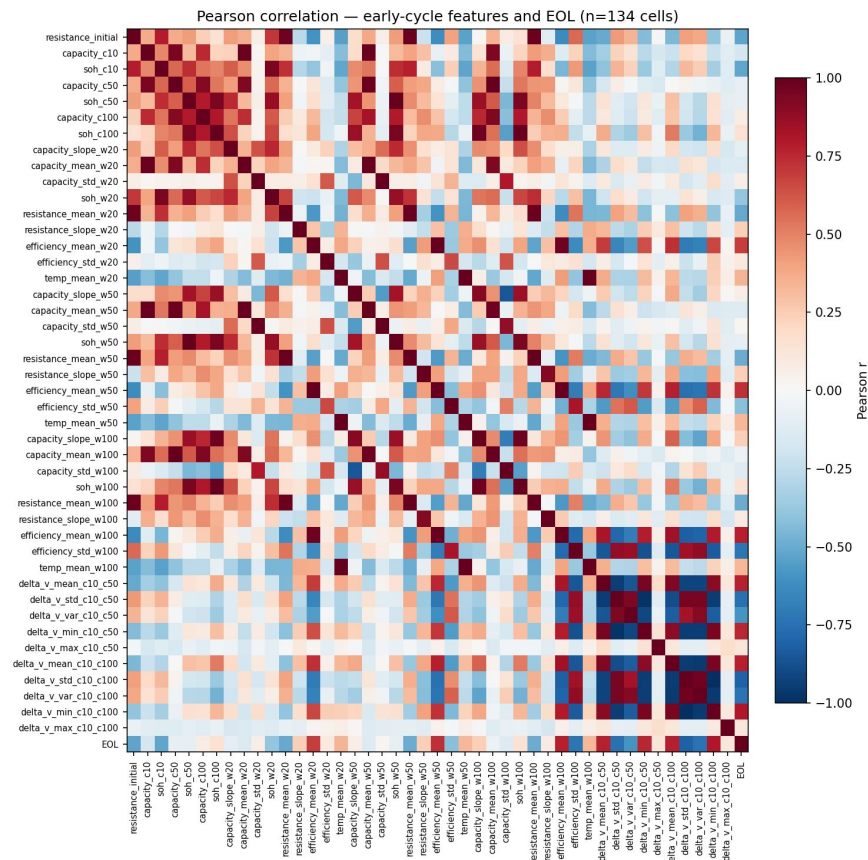


- Single cell: discharge capacity vs cycle
- Red dashed: 80% threshold · orange: computed EOL
- **Figure:**
[results/figures/capacity_fade_example.png](#)



- **134** cells · median EOL about **792** cycles · range **159–2,237**
- Wide spread motivates cell-level modeling and cautious test metrics
- **Figure:**
[results/figures/eol_distribution.png](#)

Early-cycle features (Week 3)

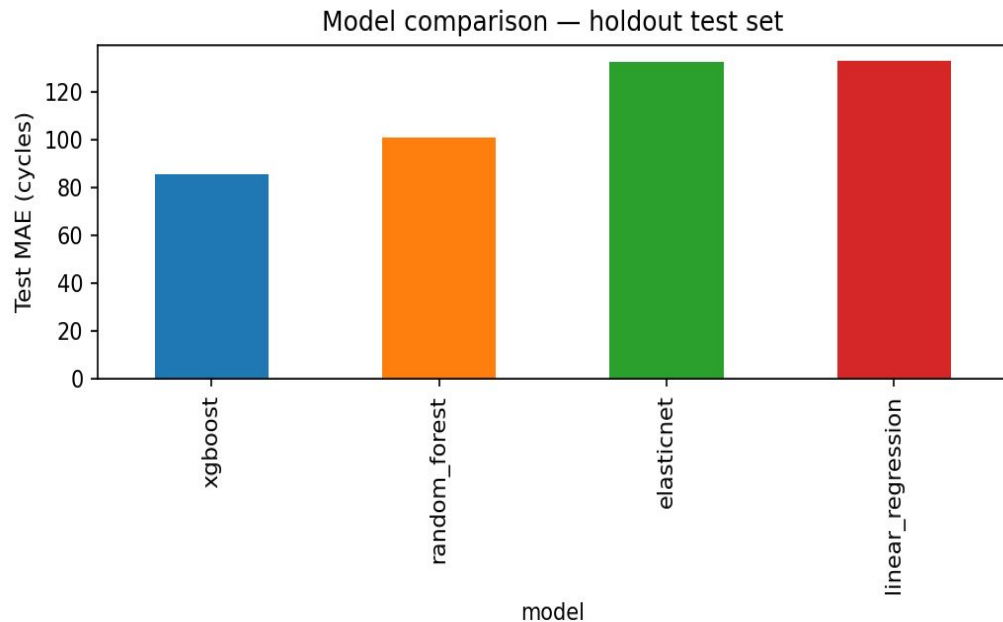


- **Input windows:** first 20 / 50 / 100 cycles per cell
- **44 features per cell:** capacity, SOH, resistance, energy efficiency, temperature summaries + **Severson-style $\Delta V(Q)$** (voltage-curve change between cycles 10→50 and 10→100)
- **Exploratory screening:** Pearson correlation vs EOL on all 134 cells
- **Strongest linear signals:** $\Delta V(Q)$ spread (`delta_v_std_c10_c100`, $|r|$ about 0.8) · efficiency (`efficiency_mean_w100`, r about 0.78)
- **Weaker alone:** raw capacity at cycle 100 — curve shape and efficiency shift before obvious fade

Figure:

results/figures/feature_correlation.png

Classical ML baselines (20-cell test set)



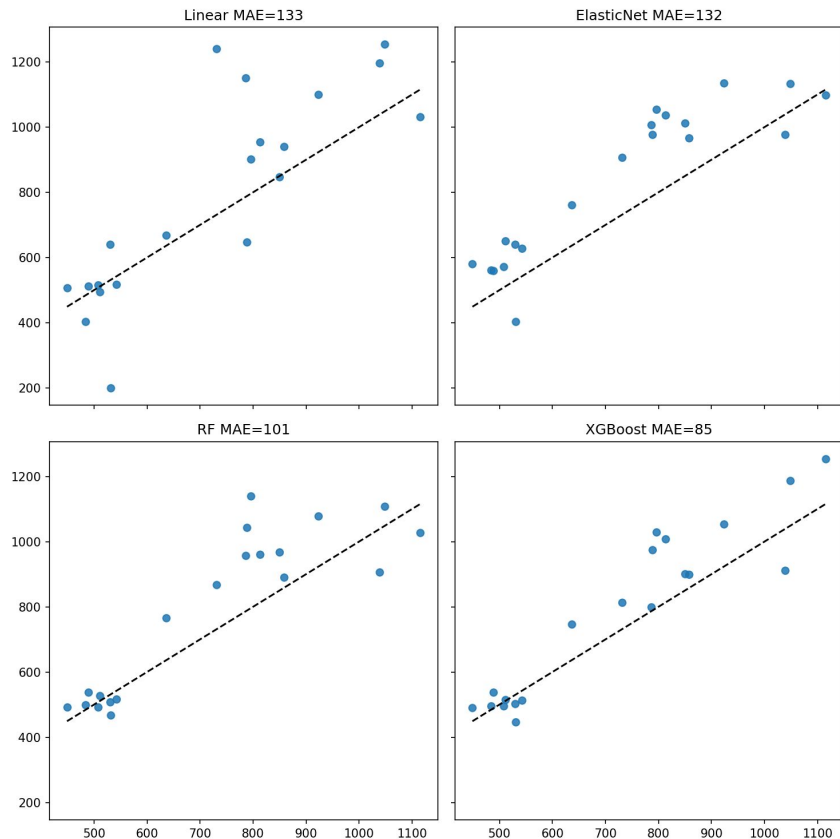
- **Input:** `cell_features.csv` — 44 features · **Target:** EOL
- **Models:** linear regression · ElasticNet · random forest · **XGBoost**
- **Tuning:** validation MAE · **Test set untouched** during hyperparameter search

Model	MAE	MAPE (about)
Linear	133	18%
ElasticNet	132	19%
Random forest	101	13%
XGBoost	85	11%

Figure:

`results/figures/model_comparison_base
lines.png`

Tabular models — predicted vs true EOL (test)

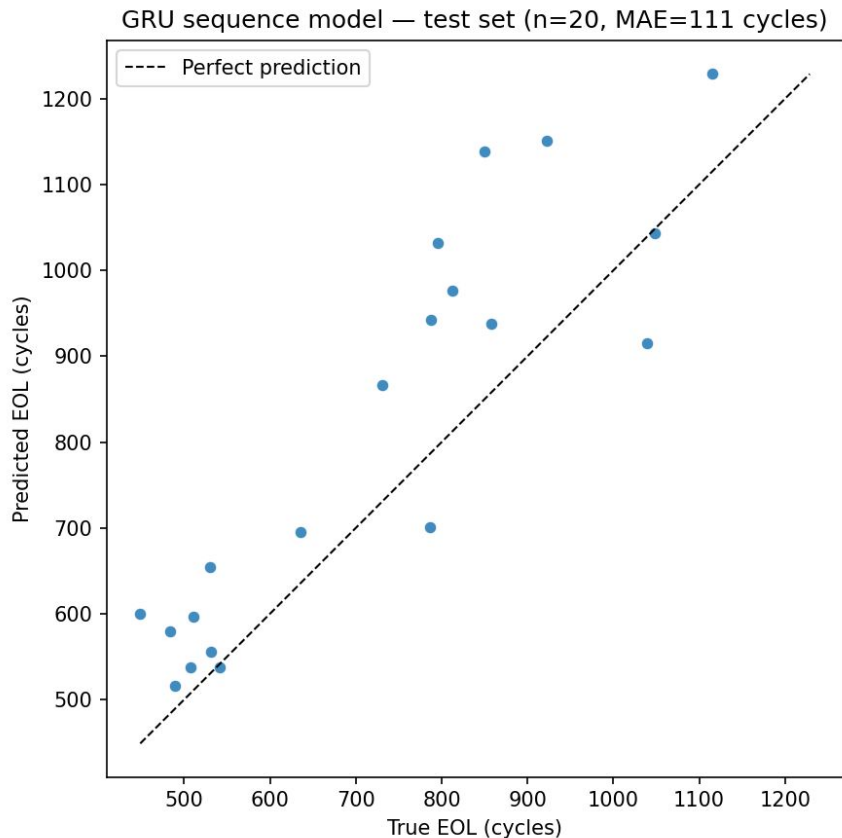


- Tree ensembles clearly beat linear models on holdout
- **XGBoost best:** about **85 cycles** MAE (about **11% MAPE**)
- RF / XGBoost importance: **$\Delta V(Q)$** and **energy efficiency** rank highest (consistent with correlation)
- **Caveat:** only **20** test cells → indicative metrics

Figure:

[results/figures/pred_vs_true_eol_baselines.png](#)

GRU on early per-cycle trajectories



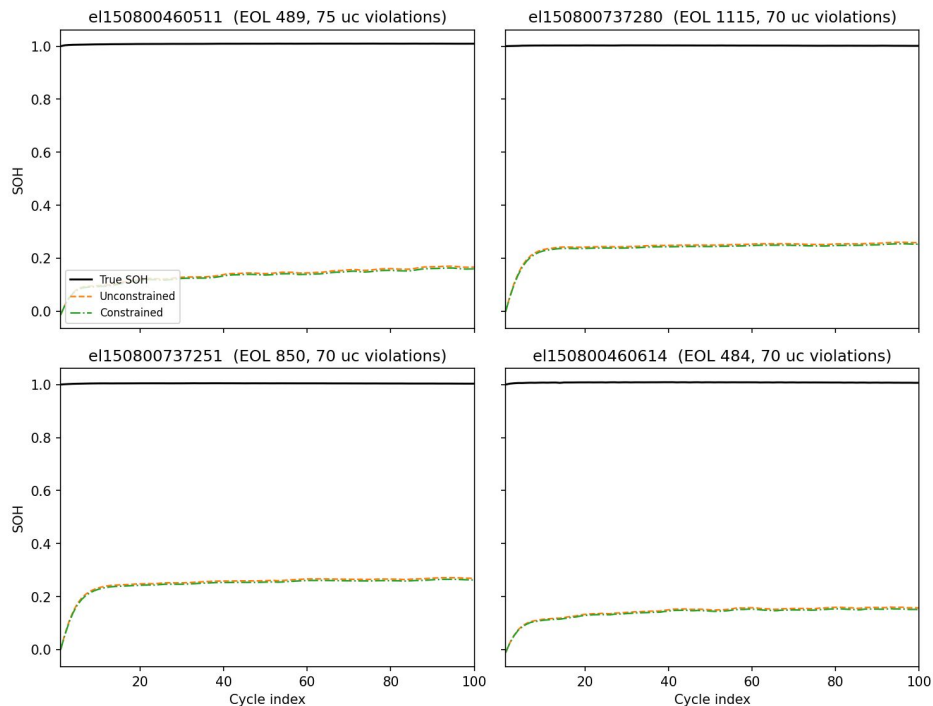
- **Input:** first **100 cycles × 4 channels** — SOH, resistance, energy efficiency, temperature ([cycle_summary.csv](#))
- **Model:** PyTorch **GRU** · 16-combo hyperparameter grid on validation MAE
- **Not in sequence input:** $\Delta V(Q)$ features (live in raw JSON only)
- **Test MAE:** about **110 cycles** — competitive but **behind XGBoost** (about 85)
- Plausible reasons: **94** training cells · no voltage-curve channels · little capacity fade in first 100 cycles

Figure:

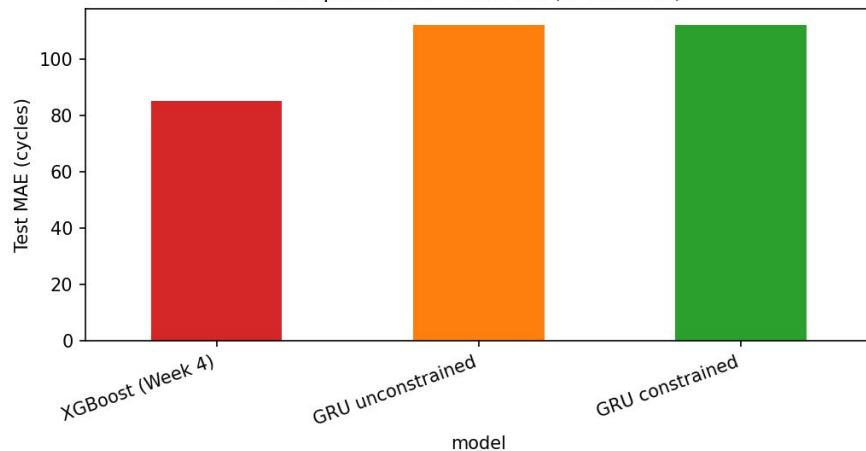
[results/figures/pred_vs_true_eol_sequence.png](#)

Physics-inspired SOH regularization (Week 6)

Example test-set SOH trajectories (cycles 1-100)



EOL prediction — test set (n=20 cells)



- **Dual-head GRU:** predicts **EOL + SOH trajectory** (100 values per cell)
- **Monotonic penalty:** discourage SOH increasing cycle-to-cycle ($\lambda \in \{0.01, 0.1, 1.0\}$)
- **EOL test MAE:** about **112 cycles** constrained vs unconstrained — **essentially unchanged**
- Violation rate drops only slightly; XGBoost still best for EOL
- Light regularizer — **not** a full physics-informed neural network

Figures: [results/figures/soh_curves_constrained.png](#),
[results/figures/model_comparison_soh_penalty.png](#)

Classical ML baselines (20-cell test set)

- **Question:** if you only observe the first N cycles, how well can you predict EOL?
- **Same split & metrics** · XGBoost uses valid feature subset at each N · GRU uses first N timesteps

N	XGBoost	GRU
20	167	144
50	107	143
100	85	110

- XGBoost gains sharply with N ($\Delta V(Q)$ from $N \geq 50$)
- GRU flat at 20–50, improves at 100
- **Best model is context-dependent** on window length

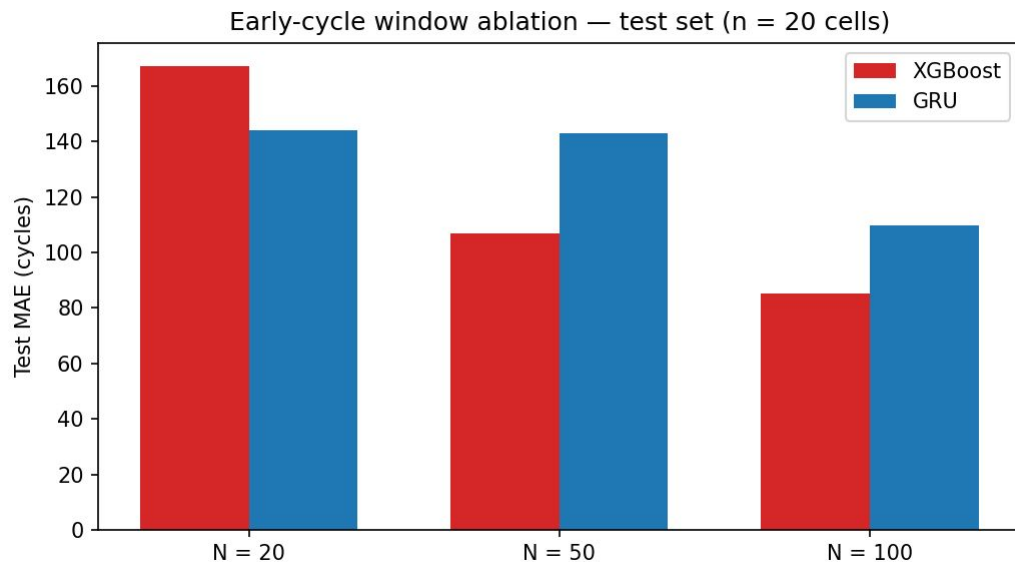


Figure:

[results/figures/ablation_early_cycles.png](#)

Limitations

- **Small cohort:** 134 cells · **20-cell** test holdout → high metric variance
- **Single dataset:** LFP/graphite · fast-charge protocol only; transfer untested
- **Asymmetric inputs:** XGBoost uses $\Delta V(Q)$ from raw JSON; GRU uses four summary channels only
- **Duplicate barcodes** in raw files (longest run kept); residual protocol heterogeneity possible
- **Mechanism-agnostic features** — capacity summaries do not identify degradation mode
- **EOL definition** — 80% of first measured capacity is one convention among many

Conclusions & future work

- Built a **reproducible pipeline** from raw JSON to ML-ready tables, models, and figures
- **Best overall:** XGBoost at $N = 100$ — test MAE about **85 cycles** (about **11% MAPE**)
- **Early screening:** about **50–100 cycles** + voltage-curve / efficiency features support useful lifetime estimates
- **Sequence model:** GRU viable but does not beat tabular model at full window on this dataset size
- **Monotonic SOH penalty:** small curve-shape effect; does not improve EOL error

Future work:

- External validation (NASA / CALCE) · $\Delta V(Q)$ channels in sequence models · larger cohorts

Deliverables: 8–12 page report · GitHub repo · [docs/future_work.md](#)